HCIA-AI V4.0 — Python & Frameworks Code Fill-in-the-Blank Guide

Official Exam Practice • 70 High-Yield Code Completion, Framework API & Math Questions (16 Points Each)

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OFFICIAL EXAM FILL-IN-THE-BLANK QUESTION FORMAT & CORE CONVENTIONS

1. Authentic Exam Testing Format
- In HCIA-AI V4.0 computer-based exams (16 Points each), fill-in-the-blank questions present natural technical statements with ____.
- You are expected to enter the exact identifier, method, or constant on the answer line without extra outer parentheses.

2. Framework Naming Traps & Case-Sensitivity
- PyTorch models subclass torch.nn.Module & use def forward(self, x):
- MindSpore models subclass mindspore.nn.Cell & use def construct(self, x):
- Class names use PascalCase (e.g. CrossEntropyLoss, Conv2d, Linear), while methods use lowercase (zero_grad, backward, step, eval).

3. Data Leakage & Evaluation Rules
- **Train Data:** calls scaler.fit_transform(X_train)
- **Test Data:** strictly calls scaler.transform(X_test) to avoid leakage.
- During evaluation, always wrap PyTorch inference inside with torch.no_grad(): and invoke model.eval().

4. Checkpoints, Math & Constants
- Official Model Files: .ckpt (Protocol Buffers checkpoint), .pt/.pth (PyTorch weights), .onnx (Cross-framework middleware).
- Core Constants: Sigmoid max derivative = **0.25**, Softmax sum = **1**, CNN Output = **floor((Input - Kernel + 2*Pad)/Stride) + 1**.

Category 1: PyTorch Deep Learning Workflow & API (Questions 1 – 15)

Question 1

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, all custom neural network models and layers must inherit from the base class torch.nn.____.

Answer: _____

Correct Answer: **Module** | **Exam Tip:** Must be capitalized: Module (PascalCase). In MindSpore, the equivalent base class is mindspore.nn.Cell.

Question 2

FILL-IN-THE-BLANK QUESTION • 16 Points

In a custom PyTorch nn.Module subclass, the computation graph executed during the forward pass must be implemented in the def ____(self, x): member function.

Answer: _____

Correct Answer: **forward** | **Exam Tip:** Must be lowercase: forward. In MindSpore, the forward method is named construct.

Question 3

FILL-IN-THE-BLANK QUESTION • 16 Points

Before computing backpropagation in PyTorch, the optimizer method called to clear previously accumulated parameter gradients is optimizer.____().

Answer: _____

Correct Answer: **zero_grad (or zero_grad())** | **Exam Tip:** Clears parameter gradients prior to computing current batch gradients to prevent gradient accumulation.

Question 4

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method invoked on the computed scalar loss tensor to calculate gradients via automatic differentiation is loss.____().

Answer: _____

Correct Answer: **backward (or backward())** | **Exam Tip:** Applies the chain rule and stores gradients into each leaf tensor's .grad attribute.

Question 5

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the optimizer method called to update neural network weights and biases according to computed gradients is optimizer.____().

Answer: _____

Correct Answer: step (or step()) | **Exam Tip:** Updates parameters in-place using the specified optimization algorithm formula (e.g., SGD, Adam).

Question 6

FILL-IN-THE-BLANK QUESTION • 16 Points

During PyTorch model evaluation and testing, the autograd engine is temporarily deactivated to conserve memory using the context manager with torch.____():.

Answer: _____

Correct Answer: no_grad (or no_grad()) | **Exam Tip:** Disables gradient tracking entirely, speeding up inference and preventing GPU out-of-memory crashes.

Question 7

FILL-IN-THE-BLANK QUESTION • 16 Points

The PyTorch nn.Module method called to switch operational behavior of layers like Dropout and BatchNorm into testing mode is model.____().

Answer: _____

Correct Answer: eval (or eval()) | **Exam Tip:** Sets self.training = False; disables dropout random zeroing and fixes BatchNorm running statistics.

Question 8

FILL-IN-THE-BLANK QUESTION • 16 Points

When implementing a custom dataset class in PyTorch, the custom class must inherit from the base class torch.utils.data.____.

Answer: _____

Correct Answer: Dataset | **Exam Tip:** Subclasses must implement __len__() and __getitem__() to support indexing and DataLoader wrapping.

Question 9

FILL-IN-THE-BLANK QUESTION • 16 Points

When initializing a PyTorch DataLoader for the training set, the boolean parameter that ensures samples are randomly reordered at each epoch is ____=True.

Answer: _____

Correct Answer: shuffle | **Exam Tip:** Generates a random permutation of dataset indices at each epoch to prevent learning mini-batch order bias.

Question 10

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method used to extract a standard Python numeric float value from a 1-element 0-dimensional tensor (e.g., scalar loss) is loss.____().

Answer: _____

Correct Answer: item (or item()) | **Exam Tip:** Prevents memory leaks from retaining graph history by extracting pure Python primitive values.

Question 11

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the core package responsible for automatic differentiation and gradient tracking is torch.____.

Answer: _____

Correct Answer: autograd | **Exam Tip:** The autograd engine provides automatic differentiation for all operations on tensors.

Question 12

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the constructor used to define a 2D spatial convolution operation over multi-channel image tensors is nn.____.

Answer: _____

Correct Answer: Conv2d | **Exam Tip:** PascalCase: nn.Conv2d. In MindSpore, the equivalent class is nn.Conv2d as well.

Question 13

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the module class that implements an affine linear transformation (matrix multiplication plus bias: y = xA^T + b) is nn.____.

Answer: _____

Correct Answer: Linear | **Exam Tip:** Dense/fully connected layer. In MindSpore, the equivalent fully connected layer class is nn.Dense.

Question 14

FILL-IN-THE-BLANK QUESTION • 16 Points

The built-in PyTorch loss function that combines LogSoftmax and Negative Log-Likelihood (NLLLoss) into a single numerically stable class is nn.____.

Answer: _____

Correct Answer: CrossEntropyLoss | **Exam Tip:** Applies LogSoftmax internally; raw unnormalized logits must be passed directly without manual Softmax.

Question 15

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the dictionary-like object returned by a model containing all learnable parameter weights and persistent buffers is model.____().

Answer: _____

Correct Answer: state_dict (or state_dict()) | **Exam Tip:** Preferred for checkpoint serialization via torch.save(model.state_dict(), 'weights.pth').

Category 2: Huawei MindSpore Framework & Ascend Platform (Questions 16 – 30)

Question 16

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, all custom neural network models and custom layers must inherit from the base class mindspore.nn.____.

Answer: _____

Correct Answer: Cell | **Exam Tip:** Fundamental building block in MindSpore (analogous to PyTorch's nn.Module).

Question 17

FILL-IN-THE-BLANK QUESTION • 16 Points

Inside a custom MindSpore nn.Cell subclass, the computational execution graph for the forward pass must be defined in the def ____(self, x): method.

Answer: _____

Correct Answer: construct | **Exam Tip:** MindSpore uses construct (PyTorch uses forward). Both implement forward propagation logic.

Question 18

FILL-IN-THE-BLANK QUESTION • 16 Points

In MindSpore, the compilation mode that constructs a static computation graph for maximum execution efficiency on Ascend hardware is ms.____.

Answer: _____

Correct Answer: GRAPH_MODE | **Exam Tip:** Static graph compilation optimizes operators on Ascend/GPU; PYNATIVE_MODE enables dynamic execution.

Question 19

FILL-IN-THE-BLANK QUESTION • 16 Points

In MindSpore, the interactive execution mode that executes operators immediately line-by-line for easy debugging is ms.____.

Answer: _____

Correct Answer: **PYNATIVE_MODE** | **Exam Tip:** Dynamic eager execution mode in MindSpore, ideal for Python debugging and research prototyping.

Question 20

FILL-IN-THE-BLANK QUESTION • 16 Points

In mindspore.set_context(), the parameter used to specify the execution hardware platform (Ascend, GPU, or CPU) is named ____.

Answer: _____

Correct Answer: **device_target** | **Exam Tip:** Accepts 'Ascend', 'GPU', or 'CPU', routing operator execution to the designated hardware accelerator.

Question 21

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, the neural network layer class that implements a fully connected feedforward linear transformation is nn.____.

Answer: _____

Correct Answer: **Dense** | **Exam Tip:** MindSpore designates fully connected layers as nn.Dense (PyTorch designates them as nn.Linear).

Question 22

FILL-IN-THE-BLANK QUESTION • 16 Points

The top-level MindSpore function used to serialize neural network parameters into a checkpoint file on disk is ms.____.

Answer: _____

Correct Answer: **save_checkpoint** | **Exam Tip:** Writes Cell parameters to disk in Huawei's native checkpoint format (e.g. lenet.ckpt).

Question 23

FILL-IN-THE-BLANK QUESTION • 16 Points

The MindSpore function used to read a serialized .ckpt file from disk and parse it into a parameter dictionary is ms.____.

Answer: _____

Correct Answer: **load_checkpoint** | **Exam Tip:** Reads the serialized checkpoint file and returns a dictionary mapping parameter names to tensors.

Question 24

FILL-IN-THE-BLANK QUESTION • 16 Points

After loading a parameter dictionary in MindSpore, the function used to load those parameters directly into the live neural network Cell is ms.____.

Answer: _____

Correct Answer: **load_param_into_net** | **Exam Tip:** Matches parameter dictionary keys with target Cell layer weights and updates model weights in-place.

Question 25

FILL-IN-THE-BLANK QUESTION • 16 Points

In modern MindSpore, the function used to create a gradient computation callable that returns both forward loss and parameter gradients is ms.____.

Answer: _____

Correct Answer: **value_and_grad** | **Exam Tip:** Creates a callable returning (loss, grads), streamlining gradient updates in PyNative mode.

Question 26

FILL-IN-THE-BLANK QUESTION • 16 Points

When preprocessing image datasets in mindspore.dataset, the operation that applies transformation pipelines across dataset columns is dataset.____().

Answer: _____

Correct Answer: map | **Exam Tip:** Executes transformation pipelines (e.g. Resize, ToTensor, Normalize) across specified columns.

Question 27

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's AI processor series specifically designed for low-power edge inference (consuming 8W) is named Ascend ____.

Answer: _____

Correct Answer: 310 | **Exam Tip:** Ascend 310 is dedicated to Edge Inference (8W); Ascend 910 is dedicated to Cloud Training (310W).

Question 28

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's flagship AI processor series designed for ultra-high-throughput cloud training (consuming 310W) is named Ascend ____.

Answer: _____

Correct Answer: 910 | **Exam Tip:** Ascend 910 delivers half-precision (FP16) compute for deep neural network cloud training.

Question 29

FILL-IN-THE-BLANK QUESTION • 16 Points

The chip enablement software architecture layer that bridges Huawei Ascend hardware with AI frameworks is called CANN, which stands for Compute Architecture for ____ Networks.

Answer: _____

Correct Answer: Neural | **Exam Tip:** CANN (Compute Architecture for Neural Networks) provides AscendCL APIs and TBE operator acceleration.

Question 30

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei Cloud, the end-to-end one-stop AI development and lifecycle management service is named ____.

Answer: _____

Correct Answer: ModelArts | **Exam Tip:** ModelArts manages data preparation, model training, hyperparameter tuning, and API inference deployment.

Category 3: Scikit-Learn Traditional Machine Learning (Questions 31 – 45)

Question 31

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, the standard function from sklearn.model_selection used to randomly partition data into training and testing sets is ____.

Answer: _____

Correct Answer: train_test_split | **Exam Tip:** Randomly partitions feature matrices X and labels y while preserving sample index alignment.

Question 32

FILL-IN-THE-BLANK QUESTION • 16 Points

In the universal Scikit-Learn Estimator API, the core method called to train an algorithm on feature matrix X and target labels y is model.____(X, y).

Answer: _____

Correct Answer: fit | **Exam Tip:** All Scikit-Learn estimators implement .fit() to calculate and store learned internal parameters.

Question 33

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, after fitting an estimator, the method used to generate discrete predicted class labels for new unseen samples is model.____(X_test).

Answer: _____

Correct Answer: predict | **Exam Tip:** Returns categorical class predictions. For class probabilities, predict_proba is used.

Question 34

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn classification estimators (such as LogisticRegression), the method that returns calibrated class probability estimates is model.____(X_test).

Answer: _____

Correct Answer: predict_proba | **Exam Tip:** Returns an array of shape (n_samples, n_classes) containing probabilities summing to 1.0 per row.

Question 35

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn preprocessing (e.g., StandardScaler), the combined method that calculates statistics on training data and applies scaling in a single step is scaler.____(X_train).

Answer: _____

Correct Answer: fit_transform | **Exam Tip:** Computes mean/std and transforms features simultaneously. Strictly applied to training data only.

Question 36

FILL-IN-THE-BLANK QUESTION • 16 Points

To prevent data leakage, after fitting a scaler on training data, the method that must be strictly called on test data is scaler.____(X_test).

Answer: _____

Correct Answer: transform | **Exam Tip:** Applies pre-computed training mean and variance to test data without recalculating parameters.

Question 37

FILL-IN-THE-BLANK QUESTION • 16 Points

In linear regression models in scikit-learn, the learned weight coefficients attribute ends with an underscore and is named model.____.

Answer: _____

Correct Answer: coef_ | **Exam Tip:** In Scikit-Learn conventions, all attributes learned during .fit() strictly end with an underscore.

Question 38

FILL-IN-THE-BLANK QUESTION • 16 Points

In linear regression models in scikit-learn, the learned bias/constant intercept attribute is named model.____.

Answer: _____

Correct Answer: intercept_ | **Exam Tip:** Represents the y-axis intercept (b); ends with an underscore as a learned attribute.

Question 39

FILL-IN-THE-BLANK QUESTION • 16 Points

In sklearn.metrics, the evaluation function that calculates the subset classification accuracy score (TP + TN) / (TP + TN + FP + FN) is ____.

Answer: _____

Correct Answer: accuracy_score | **Exam Tip:** Measures the exact match ratio between true ground-truth labels and predicted labels.

Question 40

FILL-IN-THE-BLANK QUESTION • 16 Points

In sklearn.metrics, the function that evaluates classification performance by constructing a contingency table of actual vs. predicted labels is ____.

Answer: _____

Correct Answer: confusion_matrix | **Exam Tip:** Returns a 2D matrix where rows represent actual classes and columns represent model predictions.

Question 41

FILL-IN-THE-BLANK QUESTION • 16 Points

The non-parametric supervised classification class in sklearn.tree that recursively splits nodes based on Gini impurity or Information Gain is ____.

Answer: _____

Correct Answer: DecisionTreeClassifier | **Exam Tip:** Builds binary decision trees by evaluating split criteria (Gini Impurity by default or Entropy).

Question 42

FILL-IN-THE-BLANK QUESTION • 16 Points

The ensemble meta-estimator in sklearn.ensemble that trains multiple decision trees in parallel on bootstrap samples to reduce model variance (Bagging) is ____.

Answer: _____

Correct Answer: RandomForestClassifier | **Exam Tip:** Bagging (Random Forest) reduces variance by training parallel trees on bootstrap samples.

Question 43

FILL-IN-THE-BLANK QUESTION • 16 Points

The ensemble boosting algorithm in sklearn.ensemble that trains decision trees sequentially where subsequent models correct errors of prior models to reduce bias is ____.

Answer: _____

Correct Answer: AdaBoostClassifier (or GradientBoostingClassifier) | **Exam Tip:** Boosting reduces bias by sequentially adjusting sample weights based on prior prediction errors.

Question 44

FILL-IN-THE-BLANK QUESTION • 16 Points

The unsupervised clustering algorithm in sklearn.cluster that partitions samples into k distinct clusters by minimizing within-cluster distance (inertia) is ____.

Answer: _____

Correct Answer: KMeans | **Exam Tip:** Iteratively assigns samples to closest centroids and recalculates centroids until convergence.

Question 45

FILL-IN-THE-BLANK QUESTION • 16 Points

In a trained Scikit-Learn KMeans clustering model, the attribute storing the final cluster centroid coordinates in feature space is kmeans.____.

Answer: _____

Correct Answer: cluster_centers_ | **Exam Tip:** Learned attribute ending with an underscore; stores array of shape (k, n_features).

Category 4: NumPy & Tensor Algebraic Foundations (Questions 46 – 55)

Question 46

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the ndarray method that returns a new view of the array containing the exact same data organized into new dimensions is arr.____().

Answer: _____

Correct Answer: reshape | **Exam Tip:** Changes dimensional coordinates without copying memory; passing -1 infers dimension size.

Question 47

FILL-IN-THE-BLANK QUESTION • 16 Points

The NumPy function used to insert a new singleton dimension (size 1) at a specified axis position (e.g., adding batch dimension) is np.____.

Answer: _____

Correct Answer: expand_dims | **Exam Tip:** Inserts a length-1 axis; e.g. np.expand_dims(img, axis=0) turns (28, 28) into (1, 28, 28).

Question 48

FILL-IN-THE-BLANK QUESTION • 16 Points

The NumPy function used to strip all single-dimensional entries (dimensions of length 1) from an array shape is np.____.

Answer: _____

Correct Answer: squeeze | **Exam Tip:** Removes singleton dimensions; e.g. converts (1, 28, 28, 1) into (28, 28).

Question 49

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, linear algebraic matrix multiplication between two 2D arrays (equivalent to the @ operator) is computed using the function np.____.

Answer: _____

Correct Answer: dot (or matmul) | **Exam Tip:** Inner dimension must match: (M, K) dot (K, N) yields (M, N).

Question 50

FILL-IN-THE-BLANK QUESTION • 16 Points

The NumPy function that returns the indices of the maximum values along a specified axis (used to decode softmax logits to class predictions) is np.____.

Answer: _____

Correct Answer: argmax | **Exam Tip:** Decodes prediction vectors; e.g. np.argmax(probabilities, axis=1) returns predicted class IDs.

Question 51

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy and PyTorch, the attribute used to compute the matrix transpose (swapping rows and columns) of a 2D array A is A.____.

Answer: _____

Correct Answer: T | **Exam Tip:** Property returning array with axes reversed: (A @ B).T = B.T @ A.T.

Question 52

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the function used to join a sequence of existing arrays along an existing axis is np.____.

Answer: _____

Correct Answer: concatenate | **Exam Tip:** Concatenates arrays along existing dimensions (unlike np.stack, which introduces a new axis).

Question 53

FILL-IN-THE-BLANK QUESTION • 16 Points

The NumPy function used to create a new array of a given shape and data type filled entirely with numerical zeros is np.____.

Answer: _____

Correct Answer: zeros | **Exam Tip:** Initializes empty parameter or mask arrays, e.g. np.zeros((32, 10)).

Question 54

FILL-IN-THE-BLANK QUESTION • 16 Points

The NumPy function that returns an ndarray with evenly spaced values within a given half-open interval [start, stop) is np.____.

Answer: _____

Correct Answer: arange | **Exam Tip:** Similar to Python range(), but returns an ndarray; e.g. np.arange(10).

Question 55

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy aggregation and reduction functions (such as sum, mean, or argmax), the parameter that specifies the dimension along which reduction occurs is ____.

Answer: _____

Correct Answer: axis | **Exam Tip:** axis=0 operates down columns across rows; axis=1 operates across columns along rows.

Category 5: Deep Learning Dimensions, Formulas & Constants (Questions 56 – 70)

Question 56

FILL-IN-THE-BLANK QUESTION • 16 Points

Given input feature map size I, convolution kernel size K, padding P, and stride S, the output spatial dimension formula is: Output = floor((I - K + 2*P) / S) + ____.

Answer: _____

Correct Answer: 1 | **Exam Tip:** Standard spatial formula: floor((Input - Kernel + 2*Padding) / Stride) + 1.

Question 57

FILL-IN-THE-BLANK QUESTION • 16 Points

When performing 2D convolution on an input feature map of shape (Height, Width, 64), each individual convolution kernel must have a channel depth of exactly ____.

Answer: _____

Correct Answer: 64 | **Exam Tip:** A convolution kernel's channel depth ALWAYS matches the input map's channel depth, independent of filter count.

Question 58

FILL-IN-THE-BLANK QUESTION • 16 Points

For the standard Sigmoid activation function sigma(x) = 1 / (1 + e^(-x)), the maximum value of its derivative sigma'(x) at x = 0 is exactly ____.

Answer: _____

Correct Answer: 0.25 (or 1/4) | **Exam Tip:** At x=0, sigma(0)=0.5, so max derivative is 0.5 * (1 - 0.5) = 0.25 (causing vanishing gradients in deep networks).

Question 59

FILL-IN-THE-BLANK QUESTION • 16 Points

The total sum of all output class probabilities generated by a Softmax activation layer across all classes equals exactly ____.

Answer: _____

Correct Answer: 1 (or 1.0) | **Exam Tip:** Softmax normalizes unconstrained real-valued logits into a valid probability distribution summing to 1.0.

Question 60

FILL-IN-THE-BLANK QUESTION • 16 Points

The mathematical formula for the Rectified Linear Unit (ReLU) activation function applied to input x is: f(x) = max(____, x).

Answer: _____

Correct Answer: 0 | **Exam Tip:** ReLU returns 0 for negative inputs and x for positive inputs, preventing vanishing gradients for positive activations.

Question 61

FILL-IN-THE-BLANK QUESTION • 16 Points

A Feedforward Neural Network (FNN) transmits signals strictly unidirectionally with zero feedback loops, meaning its topology is a Directed Acyclic Graph (abbreviation: ____).

Answer: _____

Correct Answer: DAG | **Exam Tip:** Directed Acyclic Graph (DAG) represents unidirectional acyclic neural data flow.

Question 62

FILL-IN-THE-BLANK QUESTION • 16 Points

In a biological or mathematical artificial neuron, the scalar bias parameter 'b' physically represents the activation ____ required to fire the neuron.

Answer: _____

Correct Answer: **threshold** | **Exam Tip:** The neuron fires (+1) if the weighted sum exceeds threshold b (or Theta); otherwise it remains inactive.

Question 63

FILL-IN-THE-BLANK QUESTION • 16 Points

Alongside L1/L2 weight regularization, the primary optimization technique used to mitigate the Exploding Gradient problem by capping gradient magnitudes is Gradient ____.

Answer: _____

Correct Answer: **Clipping** | **Exam Tip:** Gradient Clipping rescales gradient vectors exceeding a defined threshold to prevent numerical instability.

Question 64

FILL-IN-THE-BLANK QUESTION • 16 Points

In deep learning optimization, penalizing large weight values by adding an L1 or L2 norm penalty to the loss function is known as weight ____.

Answer: _____

Correct Answer: **regularization (or decay)** | **Exam Tip:** Mitigates exploding gradients and reduces overfitting by discouraging excessively large model weights.

Question 65

FILL-IN-THE-BLANK QUESTION • 16 Points

In historical AI development (Minsky & Papert, 1969), single-layer perceptrons strictly failed because they can only construct linear decision boundaries and cannot solve the ____ problem.

Answer: _____

Correct Answer: **XOR** | **Exam Tip:** XOR is linearly non-separable; solving it required Multi-Layer Perceptrons (MLPs) with non-linear activations.

Question 66

FILL-IN-THE-BLANK QUESTION • 16 Points

In the Transformer Self-Attention mechanism, the dot product between query matrix Q and key matrix K requires computing the ____ of matrix K (denoted K^T).

Answer: _____

Correct Answer: **transpose** | **Exam Tip:** Attention(Q, K, V) = Softmax((Q @ K.T) / sqrt(d_k)) @ V; Q must multiply K transpose.

Question 67

FILL-IN-THE-BLANK QUESTION • 16 Points

According to the HCIA-AI V4.0 Model File specification, the checkpoint format that uses Protocol Buffers to store weights for fine-tuning or resuming training has the extension .____.

Answer: _____

Correct Answer: **ckpt** | **Exam Tip:** Official checkpoint format in Huawei MindSpore and deep learning workflows for resuming training.

Question 68

FILL-IN-THE-BLANK QUESTION • 16 Points

In the PyTorch deep learning framework, the default recommended file format extension used to save model weights and parameter dictionaries is .____.

Answer: _____

Correct Answer: **pth (or .pt)** | **Exam Tip:** Standard serialization format for PyTorch state_dict parameter dictionaries.

Question 69

FILL-IN-THE-BLANK QUESTION • 16 Points

The open-source universal cross-framework format developed to facilitate model conversion between frameworks (e.g. PyTorch to TensorRT or Ascend) has the extension .____.

Answer: _____

Correct Answer: **onnx** | **Exam Tip:** Open Neural Network Exchange (ONNX) enables seamless cross-platform deployment and inference.

Question 70

FILL-IN-THE-BLANK QUESTION • 16 Points

In Convolutional Neural Networks, a Pooling layer (such as MaxPool2d or AvgPool2d) contains exactly ____ learnable parameters.

Answer: _____

Correct Answer: **0 (or zero)** | **Exam Tip:** Pooling performs fixed deterministic downsampling with zero learnable weights or biases.

HCIA-AI V4.0 Code Mastery: Master Answer Key & Quick Reference (All 70 Questions)

Q#	Category	Target Concept / API Context	Correct Token / Value
Q1	PyTorch Workflow & API	In PyTorch, all custom neural network models and layers...	**Module**
Q2	PyTorch Workflow & API	In a custom PyTorch nn.Module subclass, the computation...	**forward**
Q3	PyTorch Workflow & API	Before computing backpropagation in PyTorch, the optimi...	**zero_grad (or zero_grad())**
Q4	PyTorch Workflow & API	In PyTorch, the method invoked on the computed scalar l...	**backward (or backward())**
Q5	PyTorch Workflow & API	In PyTorch, the optimizer method called to update neura...	**step (or step())**
Q6	PyTorch Workflow & API	During PyTorch model evaluation and testing, the autogr...	**no_grad (or no_grad())**
Q7	PyTorch Workflow & API	The PyTorch nn.Module method called to switch operation...	**eval (or eval())**
Q8	PyTorch Workflow & API	When implementing a custom dataset class in PyTorch, th...	**Dataset**
Q9	PyTorch Workflow & API	When initializing a PyTorch DataLoader for the training...	**shuffle**
Q10	PyTorch Workflow & API	In PyTorch, the method used to extract a standard Pytho...	**item (or item())**
Q11	PyTorch Workflow & API	In PyTorch, the core package responsible for automatic ...	**autograd**
Q12	PyTorch Workflow & API	In PyTorch, the constructor used to define a 2D spatial...	**Conv2d**
Q13	PyTorch Workflow & API	In PyTorch, the module class that implements an affine ...	**Linear**
Q14	PyTorch Workflow & API	The built-in PyTorch loss function that combines LogSof...	**CrossEntropyLoss**
Q15	PyTorch Workflow & API	In PyTorch, the dictionary-like object returned by a mo...	**state_dict (or state_dict())**
Q16	MindSpore & Ascend	In Huawei MindSpore, all custom neural network models a...	**Cell**
Q17	MindSpore & Ascend	Inside a custom MindSpore nn.Cell subclass, the computa...	**construct**
Q18	MindSpore & Ascend	In MindSpore, the compilation mode that constructs a st...	**GRAPH_MODE**
Q19	MindSpore & Ascend	In MindSpore, the interactive execution mode that execu...	**PYNATIVE_MODE**
Q20	MindSpore & Ascend	In mindspore.set_context(), the parameter used to speci...	**device_target**
Q21	MindSpore & Ascend	In Huawei MindSpore, the neural network layer class tha...	**Dense**
Q22	MindSpore & Ascend	The top-level MindSpore function used to serialize neur...	**save_checkpoint**
Q23	MindSpore & Ascend	The MindSpore function used to read a serialized .ckpt ...	**load_checkpoint**
Q24	MindSpore & Ascend	After loading a parameter dictionary in MindSpore, the ...	**load_param_into_net**
Q25	MindSpore & Ascend	In modern MindSpore, the function used to create a grad...	**value_and_grad**
Q26	MindSpore & Ascend	When preprocessing image datasets in mindspore.dataset,...	**map**
Q27	MindSpore & Ascend	Huawei's AI processor series specifically designed for ...	**310**
Q28	MindSpore & Ascend	Huawei's flagship AI processor series designed for ultr...	**910**
Q29	MindSpore & Ascend	The chip enablement software architecture layer that br...	**Neural**
Q30	MindSpore & Ascend	In Huawei Cloud, the end-to-end one-stop AI development...	**ModelArts**
Q31	Scikit-Learn ML	In scikit-learn, the standard function from sklearn.mod...	**train_test_split**
Q32	Scikit-Learn ML	In the universal Scikit-Learn Estimator API, the core m...	**fit**
Q33	Scikit-Learn ML	In scikit-learn, after fitting an estimator, the method...	**predict**
Q34	Scikit-Learn ML	In scikit-learn classification estimators (such as Logi...	**predict_proba**
Q35	Scikit-Learn ML	In scikit-learn preprocessing (e.g., StandardScaler), t...	**fit_transform**
Q36	Scikit-Learn ML	To prevent data leakage, after fitting a scaler on trai...	**transform**

Q#	Category	Target Concept / API Context	Correct Token / Value
Q37	Scikit-Learn ML	In linear regression models in scikit-learn, the learne...	**coef_**
Q38	Scikit-Learn ML	In linear regression models in scikit-learn, the learne...	**intercept_**
Q39	Scikit-Learn ML	In sklearn.metrics, the evaluation function that calcul...	**accuracy_score**
Q40	Scikit-Learn ML	In sklearn.metrics, the function that evaluates classif...	**confusion_matrix**
Q41	Scikit-Learn ML	The non-parametric supervised classification class in s...	**DecisionTreeClassifier**
Q42	Scikit-Learn ML	The ensemble meta-estimator in sklearn.ensemble that tr...	**RandomForestClassifier**
Q43	Scikit-Learn ML	The ensemble boosting algorithm in sklearn.ensemble tha...	**AdaBoostClassifier (or GradientBoostingClassifier)**
Q44	Scikit-Learn ML	The unsupervised clustering algorithm in sklearn.cluste...	**KMeans**
Q45	Scikit-Learn ML	In a trained Scikit-Learn KMeans clustering model, the ...	**cluster_centers_**
Q46	NumPy & Tensor Math	In NumPy, the ndarray method that returns a new view of...	**reshape**
Q47	NumPy & Tensor Math	The NumPy function used to insert a new singleton dimen...	**expand_dims**
Q48	NumPy & Tensor Math	The NumPy function used to strip all single-dimensional...	**squeeze**
Q49	NumPy & Tensor Math	In NumPy, linear algebraic matrix multiplication betwee...	**dot (or matmul)**
Q50	NumPy & Tensor Math	The NumPy function that returns the indices of the maxi...	**argmax**
Q51	NumPy & Tensor Math	In NumPy and PyTorch, the attribute used to compute the...	**T**
Q52	NumPy & Tensor Math	In NumPy, the function used to join a sequence of exist...	**concatenate**
Q53	NumPy & Tensor Math	The NumPy function used to create a new array of a give...	**zeros**
Q54	NumPy & Tensor Math	The NumPy function that returns an ndarray with evenly ...	**arange**
Q55	NumPy & Tensor Math	In NumPy aggregation and reduction functions (such as s...	**axis**
Q56	DL Math & Constants	Given input feature map size I, convolution kernel size...	**1**
Q57	DL Math & Constants	When performing 2D convolution on an input feature map ...	**64**
Q58	DL Math & Constants	For the standard Sigmoid activation function sigma(x) =...	**0.25 (or 1/4)**
Q59	DL Math & Constants	The total sum of all output class probabilities generat...	**1 (or 1.0)**
Q60	DL Math & Constants	The mathematical formula for the Rectified Linear Unit ...	**0**
Q61	DL Math & Constants	A Feedforward Neural Network (FNN) transmits signals st...	**DAG**
Q62	DL Math & Constants	In a biological or mathematical artificial neuron, the ...	**threshold**
Q63	DL Math & Constants	Alongside L1/L2 weight regularization, the primary opti...	**Clipping**
Q64	DL Math & Constants	In deep learning optimization, penalizing large weight ...	**regularization (or decay)**
Q65	DL Math & Constants	In historical AI development (Minsky & Papert, 1969), s...	**XOR**
Q66	DL Math & Constants	In the Transformer Self-Attention mechanism, the dot pr...	**transpose**
Q67	DL Math & Constants	According to the HCIA-AI V4.0 Model File specification,...	**ckpt**
Q68	DL Math & Constants	In the PyTorch deep learning framework, the default rec...	**pth (or .pt)**
Q69	DL Math & Constants	The open-source universal cross-framework format develo...	**onnx**
Q70	DL Math & Constants	In Convolutional Neural Networks, a Pooling layer (such...	**0 (or zero)**